VersusBO: Ask an LLM Which, Not How Much

Anonymous ACL submission

Abstract

Bayesian optimization (BO) of chemical reactions spends most of its budget on categorical choices: which ligand, which base, which solvent. Large language models (LLMs) know a great deal about these reagents, yet current methods ask them for absolute quantities such as embeddings, scores or yield predictions, which are hard to calibrate and expensive to query. We measure a simpler judgment. Across five LLMs and four reaction datasets with DFT descriptors, a model orders two reagents along a named property, such as base strength, in agreement with the descriptor for 82% of pairs, while orderings implied by the same model's 1–10 ratings agree for 66%. We introduce VersusBO, which treats these pairwise comparisons as noisy evidence about named latent coordinates of each reagent. A Gaussian process models the outcome on these coordinates plus a few unnamed free dimensions, an ordered-probit likelihood with a learned reliability per axis sets how much each kind of LLM judgment counts, and VersusBO asks only the comparisons whose answers could change the next experiment. With every reliability at zero it reduces to a latent-variable GP. On nine reaction optimization tasks, VersusBO reaches the lowest final regret on seven and finds a top-5% experiment in 23% fewer iterations than the fastest baseline. When the LLM's chemistry is made wrong, VersusBO ends within 3% of a knowledge-free GP, and it uses 21 times fewer tokens than an LLM-driven optimizer.

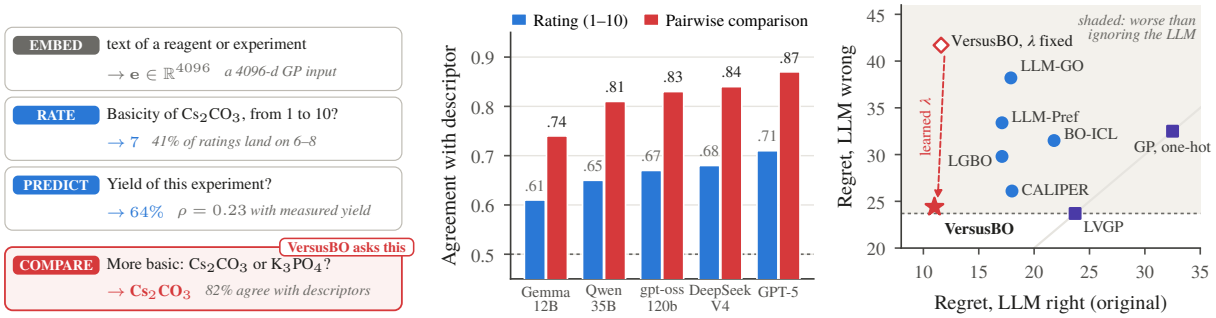
1 Introduction

A chemist optimizing a cross-coupling chooses a ligand, a base and a solvent, and sometimes a temperature and a concentration. Most of this search space is categorical, and each experiment costs hours of instrument time. Bayesian optimization (BO) handles small budgets well when its inputs carry a useful notion of distance (Shields et al., 2021). For categorical inputs that distance has to come from somewhere. One-hot encodings provide none. DFT or molecular descriptors provide one, but a chemist must choose and compute them again for every reaction class.

Large language models (LLMs) offer a second source of distance, since they have read the literature on phosphine ligands and carbonate bases. Recent work draws on that knowledge in three ways. One line embeds text descriptions of experiments and fits a Gaussian process (GP) on the embeddings, with or without finetuning the LLM (Kristiadi et al., 2024; Ranković et al., 2026). A second lets the LLM act as the surrogate model or as the optimizer itself (Liu et al., 2024; Ramos et al., 2026; MacKnight et al., 2025). A third turns LLM output into a prior over the objective (Yuan et al., 2026; Chen and Yi, 2026). All three ask the model for an absolute quantity: a vector, a predicted yield, or a score for each candidate.

Absolute quantities are where LLM output is least dependable. LLM surrogates shift their predictions with the wording and order of the prompt (Lei and Cooper, 2026), self-reported confidence can rise with error (Chen and Yi, 2026), and LLM optimizers re-propose experiments they have already run at a far higher cost per iteration than a GP (MacKnight et al., 2025). Much of what a chemist knows about reagents, by contrast, is relational: Cs_2CO_3 is a stronger base than K_2CO_3, and P(tBu)$_3$ is bulkier than PPh_3, whatever their exact descriptor values. Psychometrics has preferred paired comparisons to absolute ratings since Thurstone (1927), and work on LLM judges reports the same preference for text evaluation and ranking (Liusie et al., 2024; Qin et al., 2024).

We test whether this carries over to reagent properties. On four reaction datasets where DFT descriptors provide a ground truth, we ask five LLMs which of two reagents has more of a named property and, separately, to rate each reagent from 1 to



(a) Four ways to ask an LLM (b) Comparisons beat ratings (c) Right versus wrong oracle

Figure 1: Ask which, not how much. (a) Current methods query an LLM for embeddings, ratings or outcome predictions; VersusBO asks only for comparisons between reagents along a named property. (b) Agreement with DFT descriptor orderings on Group A, for pairwise comparisons and for orderings implied by 1–10 ratings from the same model; the dashed line marks chance. (c) Final regret on Group A when the LLM's chemistry is right (original outcomes, x-axis) and wrong (outcomes permuted across reagents, y-axis). Methods without an LLM lie on the diagonal, and the shaded region is worse than LVGP, which ignores the LLM. Learning the reliability λ moves VersusBO from the top-left to the bottom-left.

10. Comparisons agree with the descriptor ordering for 82% of pairs on average, orderings implied by the same models' ratings for 66%, and in-context yield predictions correlate with measured yields at Spearman $\rho = 0.23$ (Figure 1b). The gap appears for every model we tried.

Turning ordinal answers into geometry for BO requires a model, and that model has to stop trusting the LLM when the experiments disagree with it. We introduce VersusBO, a BO method built around that one question. It places every reagent at a point in a low-dimensional latent space (Figure 2). The LLM names a few axes, such as *ligand steric bulk*; the remaining axes stay unnamed and are fit to the data. A GP models the objective on these coordinates. An ordered-probit likelihood links each LLM answer to the coordinates through a reliability λ_k that is inferred jointly with them, so the experiments decide how much each named property counts. Between experiments, VersusBO asks only the comparisons whose answers could change the next experiment. With every λ_k at zero, VersusBO is a latent-variable GP (Zhang et al., 2020), so a misleading LLM costs at most the information its answers would have added.

Figure 1c shows what this buys. Among the LLM-based methods we evaluate, VersusBO has the lowest regret when the LLM's chemistry is right, and it is the only one that stays near a knowledge-free GP when we permute the outcomes so that the LLM's chemistry is wrong. Our contributions are (i) a measurement of comparative versus absolute LLM judgments of reagent properties across five models (§4.4); (ii) VersusBO, which infers reagent coordinates jointly from experiments and comparisons, learns a reliability per named axis, chooses comparisons by their effect on the next decision, and names new axes when unnamed dimensions start to matter (§3); (iii) an analysis showing that VersusBO contains the latent-variable GP, that its reliabilities are identifiable once named axes are standardized, and that its input dimension stays between 8 and 14 (§3.7); and (iv) an evaluation on nine reaction optimization tasks against fourteen baselines, with oracles that are made wrong, inverted or stripped of context (§4).

2 Related Work

LLMs in Bayesian optimization. Representation methods fit a GP on LLM embeddings of reaction text, either frozen (Ranković and Schwaller, 2023) or finetuned, with a Laplace approximation (Kristiadi et al., 2024) or through the GP marginal likelihood as in GOLLuM (Ranković et al., 2026; Sin et al., 2026); their GP input has 768 to 4096 dimensions. Surrogate and optimizer methods let the LLM predict outcomes or propose experiments: LLAMBO (Liu et al., 2024), BO-ICL (Ramos et al., 2026), LLM-GO (MacKnight et al., 2025), and BORA and Reasoning BO, which interleave LLM reasoning with a GP loop (Cissé et al., 2025; Yang et al., 2026). Prior methods shape the GP with LLM beliefs: LGBO shifts the GP mean toward regions the LLM prefers (Yuan et al., 2026), CALIPER gates each LLM prior by its observed

error (Chen and Yi, 2026), LABO treats LLM predictions as a cheap fidelity (Chen et al., 2026), and LLM-Pref elicits a utility from LLM preferences between reaction conditions (Patel et al., 2026). ReElicit has an LLM define interpretable features (Lin et al., 2026), and CAKE evolves GP kernels with an LLM (Suwandi et al., 2025). VersusBO asks about properties of reagents rather than outcomes, and every answer enters a likelihood whose reliability is learned from the experiments.

Categorical BO. Latent-variable GPs map each level of a categorical factor to a learned point in a low-dimensional space (Zhang et al., 2020; Oune and Bostanabad, 2021); with few observations per level, the regime of an experimental campaign, these points are poorly identified, and VersusBO adds a second likelihood that identifies them. Tabular foundation models serve as zero-shot BO surrogates (Hollmann et al., 2025; Rogers and Ponnada, 2026).

Paired comparisons. Thurstone and Bradley–Terry models explain paired comparisons through latent scale values (Thurstone, 1927; Bradley and Terry, 1952). Preference GPs and preferential BO learn a latent utility from duels (Chu and Ghahramani, 2005; González et al., 2017), and LILO converts language feedback into preferences (Kobalczyk et al., 2026). LLM judges rank text more consistently with pairwise prompts than with absolute scores (Liusie et al., 2024; Qin et al., 2024; Zheng et al., 2023), though their answers depend on option order (Wang et al., 2024). These methods compare outcomes; VersusBO compares reagents along named properties, asks in both orders, and uses the answers to shape the input space of the GP.

3 Method

3.1 Problem Setup

An experiment $x = (e_1, \dots, e_J, u)$ chooses one reagent e_j from a finite set E_j of size L_j for each categorical factor j (ligand, base, solvent) and sets numeric variables $u \in \mathbb{R}^p$ (temperature, concentration). Running x returns $y = f(x) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. The candidate pool $\mathcal{X}$ is the set of feasible combinations, and we seek its best member within T evaluations, starting from n_0 observations.

Latent-variable GPs. A latent-variable GP (LVGP) assigns each reagent $e \in E_j$ a coordinate $z_e \in \mathbb{R}^{d_j}$ and models f on the concatenated input

$$\begin{aligned} \phi_z(x) &= [z_{e_1}, \dots, z_{e_J}, u], \\ f &\sim \mathcal{GP}\big(c,\, k_\theta(\phi_z(x), \phi_z(x'))\big), \end{aligned} \tag{1}$$

where k_θ is a Matérn-5/2 kernel with automatic relevance determination. The coordinates are fit by maximizing the marginal likelihood $p(y \mid z, \theta)$ (Zhang et al., 2020). With a handful of observations per reagent many configurations of z explain the data equally well, and the GP generalizes poorly to reagents it has seen once or never.

The oracle. VersusBO sends an LLM two kinds of prompts, one naming properties of the reagents in a factor and one comparing two reagents along a named property; neither mentions the objective or an experiment (App. A).

3.2 Named and Free Coordinates

VersusBO splits each coordinate into named and free parts, $z_e = (z_e^N, z_e^F)$ with $z_e^N \in \mathbb{R}^{K_j}$ and $z_e^F \in \mathbb{R}^{R_j}$. Experiments inform all of z through the GP; LLM comparisons inform only z^N.

Naming axes. For each factor we give the LLM the reaction class, the role of the factor and the reagents by name and SMILES, and ask for up to $K_{\max} = 3$ ordered properties that a chemist would expect to affect the outcome; a second prompt merges synonyms. The name fixes the sign of each axis: a larger $z_{e,k}$ means *more* of property k. On direct arylation the oracle typically returns *steric bulk* and *σ-donor strength* for ligands and *basicity* for bases.

Identifiability and dimension. The GP likelihood is unchanged if a named axis is shifted, or stretched while its ARD length scale stretches with it, and the comparison likelihood below is unchanged under the same stretch if λ_k shrinks in proportion. We remove both symmetries by standardizing every named axis over the reagents of its factor (zero mean, unit variance, enforced by reparameterization), so λ_k is measured in units of the spread of that property. Free coordinates have prior $\mathcal{N}(0, s^2)$ with $s = 0.3$, which lets them grow only when the data ask for it, and the free coordinates of one reagent per factor are fixed at the origin. A factor with L_j levels embeds without loss in $L_j - 1$ dimensions, so we keep at most $L_j - 1$ named axes and set $R_j = \min(R,\, L_j - 1 - K_j)$. The GP input dimension is

$$d = \textstyle\sum_j (K_j + R_j) + p, \tag{2}$$

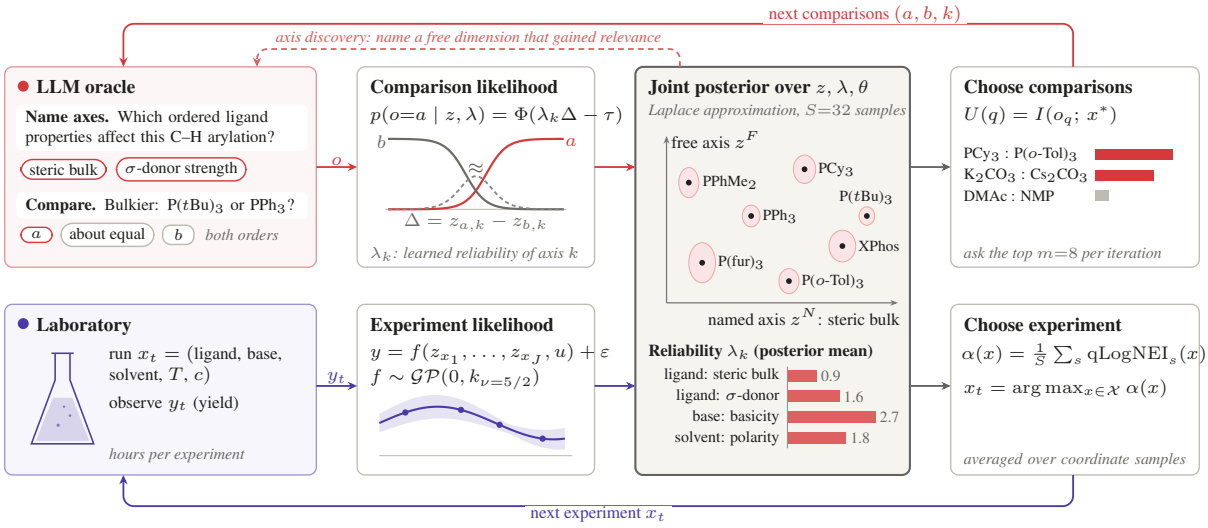


Figure 2: Overview of VersusBO. The LLM oracle (red) names ordered properties of the reagents in each factor and answers comparisons along them; the laboratory (violet) runs experiments. Each source feeds its own likelihood, and one joint posterior over reagent coordinates z, per-axis reliabilities λ and GP hyperparameters θ selects both the next comparisons, by their mutual information with the next best experiment, and the next experiment, by qLogNEI averaged over coordinate samples. Every Δ iterations, a free dimension that has gained relevance is shown to the LLM to be named (dashed). The scatter shows ligand coordinates on DA with posterior ellipses; the bars show the reliabilities learned by iteration 40.

which ranges from 8 to 14 on our tasks with $R = 2$.

3.3 Comparison Likelihood

A query $q = (a, b, k)$ asks whether reagent a has more of property k than reagent b. The answer o takes one of three values: a, b, or *about equal*. With $\Delta_q = z_{a,k} - z_{b,k}$, we use an ordered probit with tie threshold $\tau \geq 0$:

$$\begin{aligned} p(o{=}a \mid z, \lambda, \eta_q) &= \Phi(\lambda_k \Delta_q + \eta_q - \tau), \\ p(o{=}b \mid z, \lambda, \eta_q) &= \Phi(-\lambda_k \Delta_q - \eta_q - \tau), \end{aligned} \tag{3}$$

and *about equal* takes the remaining mass. The reliability $\lambda_k \geq 0$ sets how sharply the answer depends on the coordinates. At $\lambda_k = 0$ every answer is equally likely whatever z is, so comparisons along axis k carry no information. Its prior is a two-component mixture, $p(\lambda_k) = \frac{1}{2}\mathcal{HN}(\lambda_k; 0.1) + \frac{1}{2}\mathcal{HN}(\lambda_k; 3)$, a *skeptic* and a *believer*, and the data decide which component an axis belongs to.

We ask each query twice, once in each order, to cancel position bias (Wang et al., 2024). A misremembered fact misleads both phrasings, so the two answers share the query effect $\eta_q \sim \mathcal{N}(0, \sigma_\eta^2)$, which we integrate out with Gauss–Hermite quadrature; repeating a query then cannot by itself drive λ_k up. The threshold τ and scale σ_η are shared across axes and have half-normal priors.

3.4 Joint Posterior and Experiment Selection

With experiments $\mathcal{D}$ and comparisons $\mathcal{C}$, the posterior over coordinates, reliabilities and kernel hyperparameters is

$$\begin{aligned} p(z, \lambda, \theta \mid \mathcal{D}, \mathcal{C}) \propto p(y \mid z, \theta) \prod_{q \in \mathcal{C}} p(o_q \mid z, \lambda) \\ \times\, p(z)\, p(\lambda)\, p(\theta), \end{aligned} \tag{4}$$

where $p(o_q \mid z, \lambda)$ is the marginal over η_q of all answers to q, and τ, σ_η are suppressed. We fit θ by type-II maximum likelihood, find the joint mode of $(z, \log \lambda)$ with L-BFGS, and draw $S = 32$ samples from a Laplace approximation around it. The next experiment maximizes the acquisition averaged over these samples,

$$\alpha(x) = \frac{1}{S} \sum_{s=1}^{S} \text{qLogNEI}\big(\phi_{z^{(s)}}(x) \mid \mathcal{D}, \theta\big), \tag{5}$$

with qLogNEI from Ament et al. (2023). A reagent with uncertain coordinates spreads its candidates across samples, and α rewards trying it when the spread covers promising regions.

3.5 Choosing Comparisons

Most comparisons would not change what VersusBO does next. We score a candidate query by the mutual information between its answer and

the identity of the next best experiment, in the spirit of predictive entropy search (Hernández-Lobato et al., 2014). For each posterior sample s we draw a Thompson sample of f on the pool and record its maximizer $x^{*(s)}$, and we compute the answer distribution $p(o \mid z^{(s)}, \lambda^{(s)})$. The score is

$$U(q) \approx H\Big[\bar{p}(o)\Big] - \sum_{c} \frac{n_c}{S} H\Big[\bar{p}_c(o)\Big], \quad (6)$$

where $\bar{p}$ averages the answer distribution over all samples, c ranges over distinct maximizers, n_c counts the samples that chose c, and $\bar{p}_c$ averages over those samples. Candidates are pairs of reagents that appear among the top $M = 20$ experiments under α. VersusBO asks the $m = 8$ highest-scoring queries per iteration and skips the round when the best score falls below $\epsilon = 0.01$ nats.

3.6 Naming New Axes

A free dimension that the GP uses heavily marks a property the LLM did not name. Every $\Delta = 10$ iterations, VersusBO computes the relevance σ_f^2/ℓ_r^2 of each free dimension r. If it exceeds the median relevance of the named axes of that factor, VersusBO shows the LLM the three reagents with the highest and the three with the lowest posterior mean on r and asks what property separates them. The answer becomes a new named axis with the skeptic prior alone, and dimension r is reset. The six reagents shown in the naming prompt are excluded from that axis's comparisons, so the evidence that gives the new axis weight comes from reagents the prompt did not use. Algorithm 1 in App. E lists the full loop.

3.7 Properties

Proofs, full assumptions and a remark on how the input dimension enters regret bounds are in App. B.

Proposition 1 (Reduction)**.** *If $\lambda_k = 0$ for every named axis, the comparison likelihood is constant in z, and the* VersusBO *posterior over (z, θ) equals the LVGP posterior under the same prior on z.*

The statement follows from Eq. (3). Its use runs in the other direction: the skeptic component gives the posterior a path back to LVGP, so a misleading LLM costs at most the information its comparisons would have added.

Proposition 2 (Consistent reliabilities, informal)**.** *Suppose (A1) f is a GP-representable function of coordinates z^* in the model class; (A2) every reagent is tested infinitely often as T grows; and (A3) answers to distinct queries are independent given z^* and the query effects, following Eq. (3) at some $\lambda_k^* \geq 0$. Then, as T and the number of distinct queries on axis k grow, the posterior of λ_k concentrates at λ_k^*, and $\lambda_k^* = 0$ when the answers do not depend on z^* along axis k.*

Standardization makes λ_k^* unique: without it, $(c\, z_{\cdot,k},\ \lambda_k/c,\ c\,\ell_k)$ fits the data as well as $(z_{\cdot,k}, \lambda_k, \ell_k)$ for any $c > 0$. Assumption (A3) is the one most likely to fail, because LLM errors on related questions correlate; §4.3 tests the consequence.

4 Experiments

4.1 Experimental Setup

4.1.1 Benchmarks and Baselines

Tasks. We use nine pool-based tasks in three groups (Table 4, App. C). Group A carries DFT or physical-organic descriptors, which give a ground truth for the analyses in §4.4: direct arylation (DA; Shields et al., 2021), deoxyfluorination (DeoxyF; Nielsen et al., 2018), phosphoric-acid-catalysed N,S-acetal formation (NSA; Zahrt et al., 2019), and Ni-catalysed borylation (Boryl; Stevens et al., 2022) as curated by Schmid et al. (2025). Group B holds four high-throughput datasets published between 2023 and 2025 without descriptors: amide coupling (Amide, a 648-experiment subset of Zhang et al., 2025), reductive amination and Suzuki–Miyaura conversion (RedAm, SuzC; Gesmundo et al., 2023), and Chan–Lam coupling of sulfonamides (ChanLam; Gandhi et al., 2025). Group C is an emulator with 87,203 candidates built from a Ni-catalysed Suzuki campaign (NiSuz; Schlama et al., 2025). We leave out the Buchwald–Hartwig and flow Suzuki datasets (Ahneman et al., 2018; Perera et al., 2018) because earlier LLM-BO papers tuned on them.

LLMs. The main oracle is Qwen3.6-35B-A3B, an open-weight mixture-of-experts model with 3B active parameters; §4.4 also uses gpt-oss-120b, DeepSeek-V4-Flash, Gemma-4-12B and GPT-5. Every LLM-based baseline queries the main oracle, except GOLLuM, which finetunes the encoder of its original paper.

Baselines. We compare with fourteen methods in four families. *Without an LLM*: GPs on one-hot inputs and on descriptors (EDBO-style, Shields et al., 2021; DFT on Group A, PCA-reduced

RDKit elsewhere), LVGP, and TabPFN-v2 BO (Rogers and Ponnada, 2026). *LLM representations*: BoChemian, LA-PEFT (Kristiadi et al., 2024) and GOLLuM. *LLM surrogates or optimizers*: LLAMBO, BO-ICL and LLM-GO. *LLM priors or preferences*: LGBO, LLM-Pref, CALIPER and ReElicit adapted to reagents. Table 1 reports ten of them; BoChemian, LA-PEFT, LLAMBO and ReElicit trailed the strongest member of their family on every task and appear in Table 6.

4.1.2 Implementation Details

Each run starts from $n_0 = 5$ experiments drawn at random from the lower half of the outcome distribution and runs $T = 40$ sequential iterations, with 20 seeds and initial points shared across methods within a seed. GP-based methods use a Matérn-5/2 ARD kernel and qLogNEI in BoTorch (Balandat et al., 2020). VersusBO uses $K_{\max} = 3$, $R = 2$, $s = 0.3$, $S = 32$, 16 initial comparisons, $m = 8$, $M = 20$, $\epsilon = 0.01$ nats and $\Delta = 10$, with the oracle at temperature 0. We report the normalized simple regret $(y^* - y_{\text{best},T})/(y^* - y_{\text{median}}) \times 100$, the number of iterations $I_{5\%}$ until the best observed experiment lies in the top 5% of the pool (capped at 41), and the average rank across tasks. Differences are tested with paired Wilcoxon signed-rank tests over seeds and Holm correction across tasks.

4.2 Main Results

VersusBO has the lowest regret on seven of nine tasks and the second lowest on the other two (Table 1), for an average rank of 1.22 against 3.11 for LGBO, the strongest baseline. Its advantage over the runner-up is significant at $p < 0.05$ after Holm correction on DA, NSA and NiSuz and at $p < 0.1$ on Boryl and RedAm. VersusBO also finds good experiments sooner: it reaches the top 5% of the pool after 10.3 iterations on average, 23% fewer than LLM-GO (13.4), the fastest baseline, and 40% fewer than LGBO (17.3). Figure 3 shows that the separation opens within the first 10 to 15 iterations, the stage at which an LVGP has too few observations to place reagents it has rarely tested.

The two losses have identifiable causes. On DeoxyF, the descriptor GP uses DFT features that the original authors selected for this reaction, and VersusBO trails it by 1.2 points. On ChanLam, LLM-GO wins by 1.7 points. Its best conditions depend on the pairing of copper source and base, which LLM-GO reasons about directly; VersusBO recovers the relevant property only when discovery names *counter-anion basicity* for the copper source at iteration 20 (Table 11). SuzC is saturated: every method finishes within 2.4 points of the optimum, and the differences there are not significant.

Among the baselines, GOLLuM falls to tenth on RedAm, whose three numeric variables reach its encoder only as text, and LLM-GO is weak on NiSuz, where 18% of its proposals repeated an earlier experiment or fell outside the pool.

4.3 When the LLM Is Wrong

We corrupt the oracle on Group A in three ways. *Counterfactual* permutes outcomes among the levels of each factor, consistently across the pool, so the chemistry the LLM knows no longer predicts the data while the task stays equally hard without an LLM. *Inverted* flips every comparison or preference the LLM returns. *No context* strips the reaction name and literature context from every prompt, keeping SMILES only. A fourth test, *Recent*, uses Amide and ChanLam, whose best conditions the main oracle cannot name when asked (its picks fall at the 47th and 55th pool percentiles).

Under the counterfactual and inverted oracles, VersusBO ends 0.7 and 0.3 points above LVGP, within 3% (Table 2). Every other LLM-based method loses more; CALIPER, which also gates its prior on evidence, comes closest. Fixing λ removes this protection: the same model with full trust in the oracle ends at 41.7, worse than a one-hot GP (32.5). Figure 3e shows the mechanism. In counterfactual DA the posterior mean of every λ_k falls below 0.2 within 14 experiments, and VersusBO then behaves as LVGP. Removing context costs VersusBO 1.3 points and LLM-GO 4.7, so VersusBO depends less on recalling the specific system. On the recent datasets VersusBO reaches 18.1, against 18.6 for LLM-GO and 30.9 for LVGP.

4.4 Analysis

Comparisons versus ratings. On Group A we pair each named property with a descriptor that measures it (buried volume for steric bulk, pK_a of the conjugate acid for basicity, frontier-orbital energies for electronic properties; 12 pairs, Table 5). We ask which of two reagents has more of the property, in both orders; separately, we ask for a 1–10 rating of each reagent and convert ratings to orderings; and we give the model ten measured experiments in context and ask it to predict held-out yields. Averaged over five LLMs, comparisons agree with the descriptor for 82% of pairs

Method	Group A: DFT descriptors				Group B: recent HTE				C		
	DA	DeoxyF	NSA	Boryl	Amide	RedAm	SuzC	ChanLam	NiSuz	Rank ↓	$I_{5\%}$ ↓
Without an LLM											
GP, one-hot	31.4	28.7	36.2	33.5	29.8	24.1	2.1	41.3	38.6	10.89	32.1
GP, descriptors	18.2	**9.4**	22.4	24.7	21.5	19.3	1.8	35.2	27.4	7.00	22.9
LVGP	22.6	19.8	25.1	27.3	24.9	17.2	1.5	36.8	30.1	8.89	27.2
TabPFN-v2 BO	20.4	17.3	23.8	25.9	22.3	18.6	1.2	34.1	26.8	7.56	24.3
LLM representation											
GOLLuM	16.9	15.2	19.7	22.8	18.4	21.7	1.9	33.6	24.2	6.67	21.3
LLM surrogate or optimizer											
BO-ICL	19.3	18.4	26.5	23.1	17.6	20.8	2.4	30.9	29.7	8.11	21.7
LLM-GO	14.8	16.1	21.3	<u>19.2</u>	<u>12.4</u>	<u>15.9</u>	<u>0.9</u>	**24.7**	33.8	3.89	<u>13.4</u>
LLM prior or preference											
LGBO	15.6	13.4	<u>18.9</u>	20.6	16.1	16.4	1.1	29.3	<u>22.5</u>	<u>3.11</u>	17.3
LLM-Pref	<u>14.1</u>	12.7	20.2	21.4	15.3	17.8	1.4	31.6	23.9	4.33	18.0
CALIPER	15.9	14.8	19.4	21.9	16.7	16.9	1.3	30.2	23.1	4.33	18.9
VersusBO (ours)	**8.3**†	<u>10.6</u>	**11.2**†	**13.7**‡	**9.8**	**12.1**‡	**0.8**	<u>26.4</u>	**14.6**†	**1.22**	**10.3**

Table 1: Normalized simple regret ($\times 100$, lower is better) after 40 iterations, mean over 20 seeds. Rank is the average rank over the nine tasks; $I_{5\%}$ is the mean number of iterations until the best observed experiment lies in the top 5% of the pool (capped at 41). **Bold**: best; <u>underlined</u>: second best. †/‡: VersusBO is better than the runner-up at $p < 0.05$/$p < 0.1$ (paired Wilcoxon over seeds, Holm-corrected over nine tasks). Standard errors are in Table 6.

Method	Orig.	Counterf.	Invert.	No ctx.	Recent
LVGP (no LLM)	23.7	**23.7**	–	–	30.9
LLM-GO	17.9	38.2	–	22.6	18.6
BO-ICL	21.8	31.5	–	24.9	24.3
LGBO	17.1	29.8	34.6	19.8	22.7
LLM-Pref	17.1	33.4	37.9	18.4	23.5
CALIPER	18.0	26.1	25.3	19.1	23.5
VersusBO, fixed λ	11.6	41.7	45.2	**12.0**	19.4
VersusBO	**11.0**	24.4	**24.0**	12.3	**18.1**

Table 2: Regret ($\times 100$) on Group A with the original oracle and with counterfactual, inverted and context-free oracles. The last column averages Amide and ChanLam, two 2025 datasets whose best conditions the main oracle cannot name. A dash marks a test that does not apply. LVGP uses no LLM, so its counterfactual score equals its original one.

and ratings for 66%, and predictions reach Spearman $\rho = 0.23$ (Figure 1b, Table 8). The gap holds for every model and does not shrink with scale. Ratings lose accuracy mostly through ties and compressed scales: 41% of them fall on 6, 7 or 8. Averaging ten sampled ratings per reagent, which matches the number of calls in the pairwise protocol, raises their agreement only to 0.69. Acid–base comparisons reach 0.93, electronic ones 0.82 and steric ones 0.69, consistent with steric parameters being harder to read from a name or a SMILES string.

What the posterior learns. Fit on $n = 10$ random Group A experiments and evaluated on the rest of the pool, VersusBO reaches a negative log predictive density of 1.24 against 1.42 for the best baseline, and a top-decile Spearman correlation of 0.34 against 0.21 (Table 9). At $n = 20$ GOLLuM nearly closes the gap in density (1.19 against 1.08) but still trails on the top decile (0.33 against 0.42). Comparisons place reagents before the experiments can, so the gain concentrates early in a campaign.

Do the axes mean anything? For each named axis on Group A we correlate the posterior mean coordinates with the matching descriptor. The median $|\rho|$ is 0.71 for VersusBO, 0.64 when λ is fixed, so that the coordinates follow the oracle alone, and 0.36 for LVGP coordinates after an optimal rotation onto the descriptors. Experiments move the named axes toward the descriptors. Axis discovery fired on five of nine tasks (Table 11). On DA a free dimension separated the caesium carboxylates from the potassium ones; the oracle named it *cation solubility in the solvent*, and the new axis reached $\bar{\lambda} = 1.6$. On NiSuz the discovered axis never gained weight.

Cost. VersusBO asks 212 comparisons per campaign on average (424 calls), about 0.09M tokens. That is 21 times fewer than LLM-GO and 38 times fewer than LLAMBO, and the GP adds four CPU-minutes per campaign, against 1.1 GPU-hours for

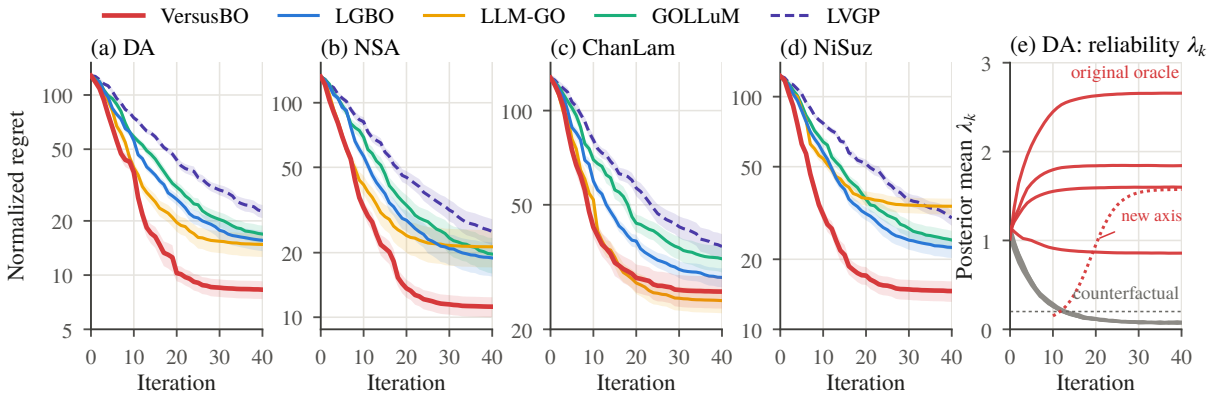


Figure 3: (a–d) Best-so-far normalized regret on four tasks (log scale; mean ± standard error over 20 seeds) for VersusBO, the two strongest baselines (LGBO, LLM-GO), the strongest representation method (GOLLuM) and LVGP, the model VersusBO reduces to; App. D shows all nine tasks. ChanLam is one of the two tasks where VersusBO is not best. (e) Posterior mean reliability λ_k of each named axis on DA (median over seeds). With the original outcomes (red) the axes keep or gain weight, and the axis named by discovery at iteration 10 (dotted) climbs to 1.6. With counterfactual outcomes (gray) every axis falls below 0.2 within 14 iterations.

Variant	All, orig.	A, counterf.
VersusBO (full)	**11.9**	24.4
A1 ratings instead of comparisons	17.2	27.9
A2 λ fixed large	12.4	41.7
A3 λ fixed at 0 (= LVGP)	22.8	23.7
A4 no free dimensions (R=0)	14.6	35.8
A5 MAP coordinates only	14.1	26.9
A6 random comparisons	15.3	24.9
A7 no axis discovery	12.8	24.6
A8 unnamed similarity queries	16.4	25.8
A9 numeric values in prompts	12.6	24.5
A10 repeats as independent	12.2	26.3

Table 3: Ablations: regret ($\times$100) averaged over all nine tasks with the original oracle, and over Group A with the counterfactual oracle (§4.3). Each row changes one component.

GOLLuM's finetuning (Table 10). With comparisons chosen at random instead of by Eq. (6), VersusBO needs 3.2 times as many to reach the same final regret.

4.5 Ablations

Each row of Table 3 changes one component. A1 tests the premise of the paper: feeding the same LLM's ratings into the same model through a Gaussian likelihood on z costs 5.3 points and leaves VersusBO level with LGBO (17.1). A2 and A3 are the two failure modes the learned reliability prevents: full trust collapses when the oracle is wrong, and zero trust gives up the gain. A4 shows that free dimensions matter most when the oracle is wrong, since they leave the GP room to place reagents where the data put them. A8 replaces named properties with unnamed similarity questions (*is a closer to b or to c?*), which carry less signal. A10 treats the two answers to a query as independent; the doubled evidence inflates λ early and costs 1.9 points under the counterfactual oracle.

Sensitivity. Across $K_{\max}$, $R \geq 1$, the number of queries per iteration, the choice of oracle, three paraphrased prompt templates and NUTS in place of the Laplace approximation, the all-task regret of VersusBO stays between 11.2 and 14.0 (App. D).

5 Conclusion

Existing methods ask LLMs for absolute numbers about experiments. We asked for comparisons between reagents along named properties and found them more accurate on every model we tried. VersusBO turns those comparisons into evidence about latent reagent coordinates, learns from the experiments how far to trust each property, and spends LLM calls only where an answer could change the next experiment. On nine reaction optimization tasks it found good conditions in fewer experiments than fourteen baselines, fell back to a knowledge-free GP when the LLM's chemistry was made wrong, and named axes that track measured physical properties. The recipe applies wherever an expensive search runs over discrete choices that a language model can describe, such as materials precursors, buffer components or software configurations.

Limitations

VersusBO needs categorical factors whose levels an LLM can recognize by name or structure; on a purely continuous problem it has nothing to ask and reduces to a GP. The number of possible comparisons grows with the square of the number of reagents. Restricting candidates to the top experiments under the acquisition keeps the number asked small, but scoring them on pools with thousands of reagents per factor will need amortization. The query effect in Eq. (3) absorbs errors shared by repeats of one question, while errors that correlate across different questions, such as a systematic misconception about a ligand class, still violate assumption (A3) of Proposition 2; the counterfactual and inverted oracles bound their cost empirically but do not model them. Axis discovery chooses what to ask from the same experiments it is later judged on; excluding the shown reagents from the new axis's comparisons removes the most direct reuse but not all of it. Finally, all campaigns are retrospective on pools of measured or emulated reactions, and a prospective campaign in a laboratory remains to be run.

Ethical Considerations

Reaction optimization could in principle speed up the synthesis of harmful compounds. VersusBO only orders reagents that a user supplies along properties found in textbooks, and it proposes no new molecules or routes, so it adds little to that risk beyond standard BO. All datasets are public. The LLM calls in this work total about 0.09M tokens per campaign, roughly one twentieth of an LLM-driven optimizer on the same task.

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A Prompts

All prompts use the chat template of the model. Fields in braces are filled per query; the system message is "You are an expert synthetic chemist. Answer briefly."

A.1 Naming axes Reaction class: {reaction}. Role of the factor: {role}.
Reagents (name; SMILES): {list}.
List up to three properties that every reagent above has, that can be ordered from less to more, and that a chemist would expect to change the outcome of this reaction. Give one short noun phrase per line.

A.2 Merging synonyms Here are property names proposed for the same reagents: {names}. Group names that describe the same property, and return one name per group.

A.3 Comparison (asked in both orders) Reaction class: {reaction}. Property: {property}.
Reagent 1: {name$_a$} ({SMILES$_a$}). Reagent 2: {name$_b$} ({SMILES$_b$}).
Which reagent has more {property}? Answer "1", "2" or "about equal".

A.4 Rating probe (§4.4 only) Reaction class: {reaction}. Property: {property}. Reagent: {name} ({SMILES}).
Rate the {property} of this reagent among typical {role}s on a scale from 1 (least) to 10 (most). Answer with one integer.

A.5 Axis discovery Reaction class: {reaction}. Role of the factor: {role}.
Group 1: {three reagents}. Group 2: {three reagents}.
Name one property, other than {existing axes}, that is high for Group 1 and low for Group 2. Give one short noun phrase.

For the *no-context* test (§4.3) we replace the reaction class with "a chemical reaction" and the role with "reagent", and drop all names, keeping SMILES only.

B Proofs

Proposition 1. With $\lambda_k = 0$, Eq. (3) gives $p(o{=}a \mid z, \eta_q) = \Phi(\eta_q - \tau)$ and $p(o{=}b \mid z, \eta_q) =$

$\Phi(-\eta_q - \tau)$, neither of which depends on z. Integrating over η_q yields a likelihood for the answers to q that depends on (τ, σ_η) only. The product over $q \in \mathcal{C}$ is therefore a constant in (z, θ), and Eq. (4) reduces to $p(y \mid z, \theta)\, p(z)\, p(\theta)$, the LVGP posterior under the same prior and parameterization of z. Because the comparison likelihood is continuous in λ, the conditional posterior over (z, θ) given λ converges to the LVGP posterior as $\lambda \to 0$, and so does the marginal posterior when the posterior of λ concentrates at zero. □

Proposition 2: assumptions. (A1) There are coordinates z^* and a function g in the RKHS of k_θ such that $f = g \circ \phi_{z^*}$; after standardizing named axes and pinning one free coordinate per factor, z^* is unique up to rotations of the free coordinates within each factor and reflections of each named axis. (A2) Every reagent is tested infinitely often as $T \to \infty$, the numeric inputs are dense in a compact set, and θ is identified from the observed pairs. (A3) Given z^* and the query effects $\{\eta_q\}$, answers to distinct queries are independent and follow Eq. (3) at some $(\lambda_k^*, \tau^*, \sigma_\eta^*)$. (A4) Each query is answered $r \geq 2$ times, and the empirical variance of the differences Δ_q queried on axis k stays bounded away from zero.

Proof sketch. The argument has two stages. *Stage 1.* Under (A1)–(A2), standard posterior consistency for GP regression with a correctly specified mean function implies that the posterior over z concentrates on the equivalence class of z^* described in (A1). Rotations of free coordinates leave both likelihoods unchanged and do not involve λ. A reflection of named axis k leaves the GP likelihood unchanged; under the constraint $\lambda_k \geq 0$ the comparison likelihood selects the orientation in which the property increases along the axis whenever $\lambda_k^* > 0$, and is invariant to it when $\lambda_k^* = 0$. *Stage 2.* For z in a neighborhood B_δ of the class of z^*, the answers on axis k follow an ordered-probit model with a random intercept per query, known covariates $\Delta_q(z)$ and unknown slope λ_k. With clusters of size $r \geq 2$ the slope, thresholds and random-effect scale are identified, and (A4) keeps the design non-degenerate, so the posterior of λ_k given $z \in B_\delta$ concentrates at λ_k^* as the number of distinct queries grows; continuity of Eq. (3) in Δ makes this uniform over B_δ up to an error that vanishes with δ. Combining the stages, $\Pr(|\lambda_k - \lambda_k^*| > \epsilon \mid \mathcal{D}, \mathcal{C}) \leq \Pr(z \notin B_\delta \mid \mathcal{D}, \mathcal{C}) + \sup_{z \in B_\delta} \Pr(|\lambda_k - \lambda_k^*| > \epsilon \mid z, \mathcal{D}, \mathcal{C})$, and both terms vanish. If the answers do not depend on z^* along axis k, the slope that generates them is zero, so $\lambda_k^* = 0$. □

Why standardization is needed. For $c > 0$, replace $z_{\cdot,k}$ by $c\, z_{\cdot,k}$, the ARD length scale ℓ_k by $c\,\ell_k$ and λ_k by λ_k / c. The kernel depends on $z_{\cdot,k}/\ell_k$ and the probit on $\lambda_k \Delta_q$, so both likelihoods are unchanged, and λ_k is not identified without fixing the scale of the axis. Unit variance over the reagents of a factor fixes it.

Remark (dimension and regret). Regret bounds for GP bandits grow with the maximal information gain of the kernel, $\tilde{O}(T^{d/(2\nu+d)})$ for Matérn-ν kernels on d inputs (Vakili et al., 2021). Eq. (2) keeps d between 8 and 14, whereas embedding-based surrogates work with d between 768 and 4096. Coordinate error ε adds a term linear in εT (Bogunovic and Krause, 2021); comparisons reduce ε for the reagents they touch, and §3.5 directs them to the reagents that matter for the next decision.

C Datasets

Sources and preprocessing. Group A tasks follow the pools of the original publications; Boryl uses the 33 aryl electrophiles, 23 ligands and 2 solvents curated by Schmid et al. (2025) from Stevens et al. (2022). Amide keeps three acids, three amines and the 72 condition sets that were run for all nine substrate pairs in Zhang et al. (2025). NiSuz is the emulator released with Schlama et al. (2025), which keeps only conditions with valid predictions; its pool is therefore not a full Cartesian product. Objectives are maximized; for NSA we maximize $\Delta\Delta G^{\ddagger}$.

Descriptor pairing for the analysis. Table 5 lists the 12 property–descriptor pairs used in §4.4. Pairs of reagents whose descriptor values differ by less than 2% of the descriptor's range are excluded as ties.

Counterfactual and inverted tests. For the counterfactual test we draw, per factor and seed, a random permutation π_j of E_j, and give each experiment the outcome of the experiment whose j-th reagent is $\pi_j(e_j)$ for every j, with the same numeric settings. This keeps the outcome distribution and the interaction structure of the pool, and a method that ignores reagent identity faces the same task. The inverted test maps every comparison answer $a \leftrightarrow b$ and, for preference-based baselines,

ID	Reaction	Objective	Pool	Categorical factors (levels)	Numeric	d	Group
DA	Direct arylation	yield	1,728	ligand (12), base (4), solvent (4)	T, conc.	12	A
DeoxyF	Deoxyfluorination	yield	740	alcohol (37), RSO_2F (5), base (4)	–	10	A
NSA	CPA N,S-acetal formation	$\Delta\Delta G^{\ddagger}$	1,075	catalyst (43), imine (5), thiol (5)	–	10	A
Boryl	Ni-cat. borylation	yield	1,518	Ar–X (33), ligand (23), solvent (2)	–	9	A
Amide	Amide coupling	yield	648	acid (3), amine (3), conditions (72)	–	8	B
RedAm	Reductive amination	conversion	768	substrate (16), solvent (2)	3	8	B
SuzC	Suzuki–Miyaura	conversion	960	Ar–X (8), boronate (10), conditions (12)	–	12	B
ChanLam	Chan–Lam coupling	wtd. selectivity	1,920	RSO_2NH_2 (10), $ArB(OH)_2$ (2), Cu (4), solvent (4), base (6)	–	14	B
NiSuz	Ni-cat. Suzuki (emulated)	yield	87,203	ligand, base, solvent, additive	–	13	C

Table 4: Benchmarks. Pool is the number of candidate experiments; d is the VersusBO input dimension (Eq. 2) for the axes named by the main oracle. NiSuz is an emulator built from 576 HTE reactions; its pool keeps only emulated conditions with valid predictions, so it is not a full Cartesian product.

Task	Factor	Named property	Descriptor	Pairwise accuracy
DA	ligand	steric bulk	$\%V_{\mathrm{bur}}$	0.66
DA	ligand	σ-donor strength	HOMO energy	0.80
DA	base	basicity	$\mathrm{p}K_{\mathrm{a}}$(H)	0.94
DA	solvent	polarity	dielectric const.	0.92
DeoxyF	alcohol	steric hindrance	$\%V_{\mathrm{bur}}$ (C_{α})	0.72
DeoxyF	sulfonyl fluoride	electrophilicity	LUMO energy	0.82
DeoxyF	base	basicity	$\mathrm{p}K_{\mathrm{a}}$(H), MeCN	0.93
NSA	catalyst	3,3′-substituent bulk	$\%V_{\mathrm{bur}}$	0.67
NSA	catalyst	Brønsted acidity	$\mathrm{p}K_{\mathrm{a}}$	0.91
NSA	imine	electrophilicity	LUMO energy	0.81
Boryl	ligand	steric bulk	$\%V_{\mathrm{bur}}$	0.70
Boryl	aryl electrophile	C–X electrophilicity	LUMO energy	0.84

Table 5: Pairwise accuracy of the main oracle for each property–descriptor pair on Group A. Pairs tied on the descriptor are excluded.

reverses every preference.

Recent split. For Amide and ChanLam we ask the main oracle, with the full reaction description, to name the five best conditions. Their mean percentile in the pool is 47 for Amide and 55 for ChanLam, close to the 50 expected by chance.

D Additional Results

Sensitivity. All-task regret for $K_{\max} \in \{1, 2, 3, 5\}$ is 14.0, 12.5, 11.9 and 12.4; for $R \in \{0, 1, 2, 4\}$ it is 14.6, 12.2, 11.9 and 12.3 ($R = 0$ is ablation A4); and for $m \in \{2, 4, 8, 16\}$ queries per iteration it is 13.4, 12.3, 11.9 and 11.8. Replacing the main oracle with Gemma-4-12B gives 13.8 and with GPT-5 gives 11.2. Across three paraphrased prompt templates, the standard deviation of the all-task mean is 0.7 for VersusBO and 3.9 for LLM-GO. NUTS in place of the Laplace approximation gives 11.6 at nine times the wall-clock time.

Batch setting. With batches of $q = 8$ and five rounds (40 experiments), the all-task mean regret is 13.0 for VersusBO, 18.2 for LGBO, 20.6 for GOLLuM and 24.1 for LVGP. VersusBO selects the batch greedily with the joint qLogNEI of Eq. (5) and asks comparisons once per round.

Cross-domain check. On the hybrid perovskite band-gap task distributed with Olympus (Häse et al., 2021), whose factors are an organic cation, a metal cation and a halide, VersusBO reaches a regret of 9.1 after 40 iterations, against 12.6 for LGBO, 13.2 for GOLLuM and 15.8 for LVGP. The oracle named *ionic radius* for both cation factors and *electronegativity* for the halide.

E Implementation

Model and inference. The kernel is Matérn-5/2 with ARD over all d inputs, a $\mathrm{Gamma}(3, 6)$ prior on length scales and a constant mean. Named coordinates are parameterized by unconstrained vectors that are centered and scaled per axis; free coordinates have prior $\mathcal{N}(0, 0.3^2)$. The tie threshold has prior $\mathcal{HN}(0.5)$ and the query-effect scale $\mathcal{HN}(1)$. The joint mode is found with L-BFGS from the

Method	DA	DeoxyF	NSA	Boryl	Amide	RedAm	SuzC	ChanLam	NiSuz
GP, one-hot	31.4 ±3.69	28.7 ±3.08	36.2 ±5.03	33.5 ±3.36	29.8 ±1.72	24.1 ±2.04	2.1 ±0.24	41.3 ±3.12	38.6 ±3.64
GP, descriptors	18.2 ±1.86	9.4 ±1.01	22.4 ±2.28	24.7 ±2.47	21.5 ±2.27	19.3 ±1.95	1.8 ±0.22	35.2 ±3.32	27.4 ±1.98
LVGP	22.6 ±1.88	19.8 ±2.40	25.1 ±3.55	27.3 ±2.93	24.9 ±2.50	17.2 ±1.82	1.5 ±0.17	36.8 ±3.49	30.1 ±2.22
TabPFN-v2 BO	20.4 ±1.92	17.3 ±2.41	23.8 ±3.09	25.9 ±2.55	22.3 ±2.82	18.6 ±1.66	1.2 ±0.14	34.1 ±4.94	26.8 ±1.94
GOLLuM	16.9 ±1.94	15.2 ±1.81	19.7 ±3.71	22.8 ±1.95	18.4 ±1.19	21.7 ±3.20	1.9 ±0.19	33.6 ±3.23	24.2 ±2.31
BO-ICL	19.3 ±2.50	18.4 ±2.07	26.5 ±3.32	23.1 ±1.96	17.6 ±1.83	20.8 ±2.31	2.4 ±0.22	30.9 ±2.97	29.7 ±2.20
LLM-GO	14.8 ±2.19	16.1 ±2.72	21.3 ±4.23	19.2 ±2.29	12.4 ±1.14	15.9 ±1.66	0.9 ±0.10	24.7 ±2.20	33.8 ±2.66
LGBO	15.6 ±2.03	13.4 ±1.50	18.9 ±3.43	20.6 ±1.83	16.1 ±1.69	16.4 ±1.49	1.1 ±0.12	29.3 ±2.02	22.5 ±2.24
LLM-Pref	14.1 ±1.53	12.7 ±1.82	20.2 ±2.00	21.4 ±2.61	15.3 ±1.25	17.8 ±1.63	1.4 ±0.15	31.6 ±3.21	23.9 ±2.07
CALIPER	15.9 ±2.23	14.8 ±1.44	19.4 ±3.02	21.9 ±2.25	16.7 ±1.35	16.9 ±1.87	1.3 ±0.13	30.2 ±2.94	23.1 ±2.31
VersusBO	8.3 ±0.94	10.6 ±0.99	11.2 ±1.20	13.7 ±1.45	9.8 ±0.76	12.1 ±1.20	0.8 ±0.07	26.4 ±3.10	14.6 ±1.48
BoChemian	19.8 ±1.82	17.9 ±2.19	23.6 ±3.18	25.4 ±3.66	20.7 ±1.70	22.9 ±2.53	2.2 ±0.26	35.1 ±3.43	26.9 ±1.95
LA-PEFT	18.4 ±2.09	16.8 ±1.78	22.1 ±2.80	24.3 ±2.82	19.9 ±2.59	23.4 ±2.24	2.0 ±0.20	34.4 ±2.87	26.1 ±2.15
LLAMBO	20.6 ±2.00	19.7 ±2.44	27.4 ±2.88	24.8 ±2.44	18.9 ±1.85	21.5 ±1.80	2.6 ±0.21	32.8 ±4.58	31.2 ±3.04
ReElicit (adapted)	18.9 ±1.64	16.3 ±1.74	21.8 ±3.41	23.9 ±2.40	19.6 ±1.91	19.4 ±1.87	1.7 ±0.22	33.0 ±4.42	25.7 ±2.02

Table 6: Mean ± standard error of the final normalized regret ($\times 100$) over 20 seeds, including the four baselines reported only here (bottom block). Average ranks over all 15 methods: BoChemian 11.33, LA-PEFT 9.67, LLAMBO 11.67, ReElicit (adapted) 8.00, VersusBO 1.22.

Method	DA	DeoxyF	NSA	Boryl	Amide	RedAm	SuzC	ChanLam	NiSuz	Mean
GP, one-hot	35.7	36.9	33.1	31.8	37.7	33.3	21.8	27.5	31.1	32.1
GP, descriptors	25.8	21.0	22.9	24.1	27.6	26.6	16.4	20.1	21.8	22.9
LVGP	31.3	31.0	27.5	27.9	30.1	27.5	17.7	24.1	28.0	27.2
TabPFN-v2 BO	28.7	28.3	25.0	26.1	26.2	26.7	16.1	20.7	21.4	24.3
GOLLuM	23.3	25.0	20.5	21.4	21.9	26.1	15.6	18.6	19.5	21.3
BO-ICL	24.4	26.2	24.1	19.9	22.3	24.5	16.8	17.0	20.5	21.7
LLM-GO	15.1	19.2	12.4	11.5	12.2	13.6	8.5	9.9	18.8	13.4
LGBO	21.3	20.1	16.3	17.4	19.3	20.0	12.9	13.6	15.2	17.3
LLM-Pref	18.3	20.9	19.5	19.9	19.0	20.2	12.3	15.6	16.7	18.0
CALIPER	21.8	23.1	18.2	19.3	19.5	22.0	12.6	16.9	16.7	18.9
VersusBO	10.8	11.7	10.3	9.7	10.5	12.8	8.6	10.4	8.2	10.3

Table 7: Iterations until the best observed experiment lies in the top 5% of the pool ($I_{5\%}$, capped at 41), mean over 20 seeds.

LLM	Pairwise	From ratings	Prediction ρ
Gemma-4-12B	0.74	0.61	0.12
Qwen3.6-35B-A3B	0.81	0.65	0.21
gpt-oss-120b	0.83	0.67	0.24
DeepSeek-V4-Flash	0.84	0.68	0.26
GPT-5	0.87	0.71	0.31
Mean	0.82	0.66	0.23

Table 8: Accuracy of LLM judgments on Group A. The first two columns give the fraction of reagent pairs ordered as the matching descriptor orders them (macro-average over 12 property–descriptor pairs); "about equal" answers and tied ratings count as half correct. The last column is the Spearman correlation between in-context yield predictions and measured outcomes on held-out experiments.

	NLPD ↓		Top-decile ρ ↑	
Surrogate	n=10	n=20	n=10	n=20
GP, descriptors	1.42	1.21	0.18	0.31
LVGP	1.58	1.30	0.09	0.24
GOLLuM	1.47	1.19	0.16	0.33
BO-ICL	1.71	1.52	0.21	0.29
VersusBO	**1.24**	**1.08**	**0.34**	**0.42**

Table 9: Surrogate quality on Group A from n random experiments, averaged over 20 splits. NLPD: negative log predictive density on the rest of the pool; top-decile ρ: Spearman correlation on the best 10% of the pool.

previous mode and from five random restarts; the Laplace covariance uses the exact Hessian of the log posterior in $(z, \log\lambda, \log\tau, \log\sigma_\eta)$, computed with automatic differentiation. One iteration takes about six seconds on 16 CPU cores, of which the query scoring takes two.

Baselines. All GP baselines share the kernel, priors and acquisition of VersusBO. LLM-based baselines use the hyperparameters of their papers and the main oracle at the decoding settings below; we tuned only the number of in-context examples for BO-ICL and LLAMBO on DA and kept it fixed elsewhere. ReElicit was designed for sys-

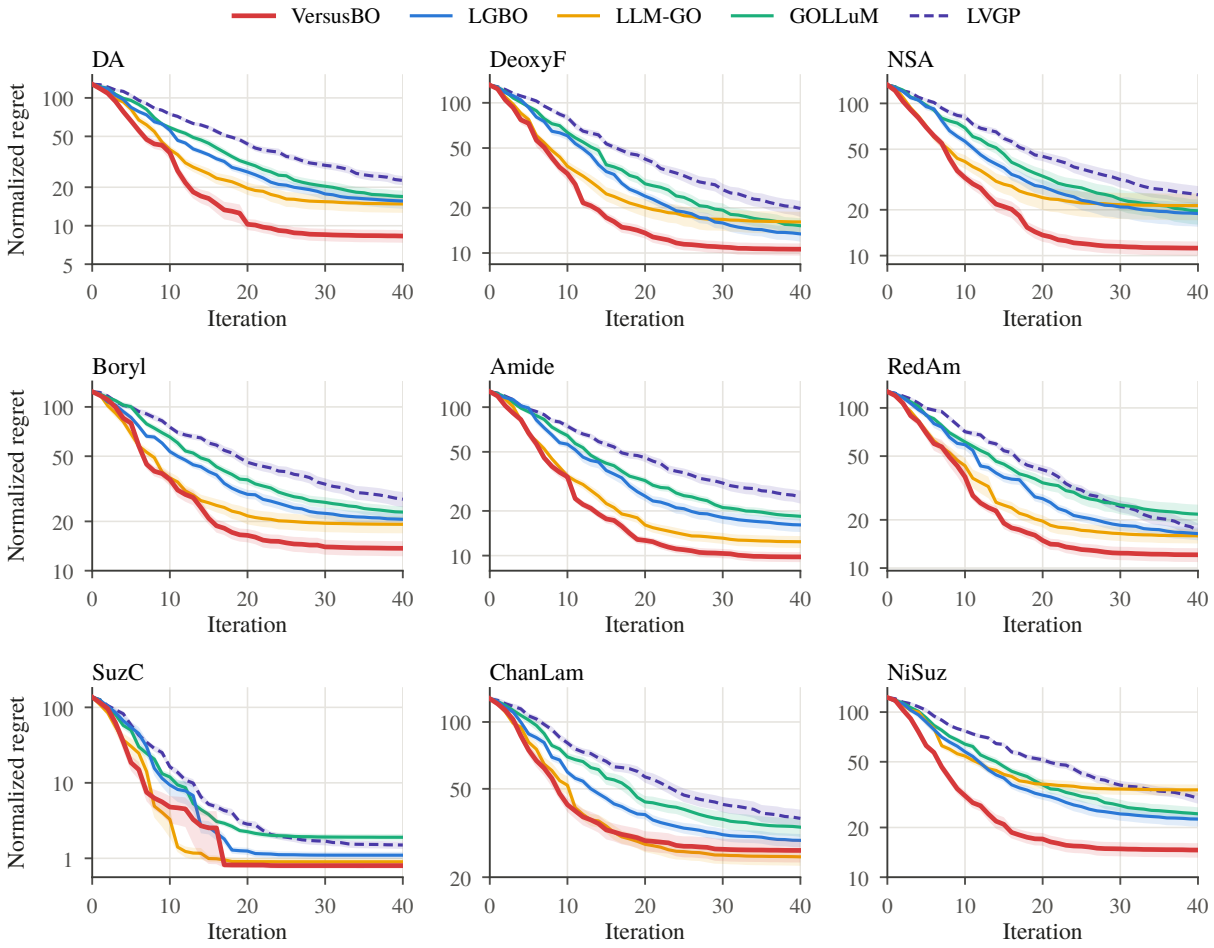


Figure 4: Best-so-far normalized regret on all nine tasks (log scale; mean ± standard error over 20 seeds).

Method	LLM tokens	Other compute
LLAMBO	3.4M	2 CPU-min
BO-ICL	2.6M	2 CPU-min
LLM-GO	1.9M	–
LGBO	0.8M	3 CPU-min
CALIPER	0.6M	3 CPU-min
LLM-Pref	0.15M	2 CPU-min
GOLLuM	–	1.1 GPU-h (H100)
VersusBO	**0.09M**	4 CPU-min

Table 10: Cost of one 40-iteration campaign on DA with the main oracle (input plus output tokens), mean over seeds. GOLLuM finetunes its own encoder and queries no generative model.

Algorithm 1 VersusBO

Require: pool $\mathcal{X}$, initial data $\mathcal{D}$, budget T, oracle
1: name axes for each factor (Prompts A.1, A.2); set $\mathcal{C} \leftarrow$ answers to 16 random comparisons
2: **for** $t = 1, \ldots, T$ **do**
3: fit Eq. (4) on $(\mathcal{D}, \mathcal{C})$; draw S Laplace samples
4: score candidate queries by Eq. (6); if the best exceeds ϵ, ask the top m in both orders, add them to $\mathcal{C}$ and refit
5: $x_t \leftarrow \arg\max_{x \in \mathcal{X} \setminus \mathcal{D}} \alpha(x)$ (Eq. (5)); run x_t; $\mathcal{D} \leftarrow \mathcal{D} \cup \{(x_t, y_t)\}$
6: **if** $t \bmod \Delta = 0$ **then** test free dimensions and name new axes (§3.6)
7: **end if**
8: **end for**
9: **return** the best observed experiment

tem prompts; we adapt it by letting the LLM define reagent features from the same naming prompt as VersusBO, with the features entering a GP as fixed inputs.

LLM versions and decoding. The open-weight models (Qwen3.6-35B-A3B, gpt-oss-120b, Gemma-4-12B) run with vLLM in bf16 on one H100. DeepSeek-V4-Flash and GPT-5 were accessed through their providers' APIs in September 2026. Naming and comparison prompts use temperature 0 and at most 16 output tokens; the budget-matched rating probe samples at temperature 0.7. Reasoning modes are disabled where a model offers the choice.

Task	Factor	Reagents shown (high vs. low)	Property named by the oracle	$\bar{\lambda}$ at t=40
DA	base	Cs^+ vs. K^+ carboxylates	cation solubility in the solvent	1.6
NSA	catalyst	extended vs. simple 3,3′-aryls	3,3′-aryl π-surface	1.2
Boryl	ligand	bidentate vs. monodentate	chelation (denticity)	0.9
ChanLam	Cu source	$Cu(OAc)_2$ vs. $Cu(OTf)_2$	counter-anion basicity	1.3
NiSuz	additive	salts vs. none	Lewis acidity of additive	0.2

Table 11: Axes added by discovery (§3.6), with the posterior mean reliability at iteration 40 averaged over the seeds where the axis was added. The NiSuz axis never gained weight.